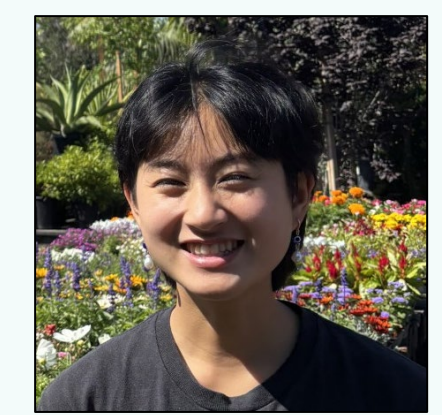




Floodplain Mixology: The Role of Tributary Signatures in Yolo Bypass Management and Restoration



Kristina.Nguyen
@water.ca.gov

Kristina Nguyen¹, Amanda Niemela¹, Spencer Breining-Aday¹, Sydney Stark², Daniel Consolvo², Soffia Ramsey², Carson Jeffres², Mackenzie Miner¹, Eric Holmes¹

¹California Department of Water Resources, ²University of California, Davis

Introduction

- The Yolo Bypass is multi-benefit landscape which supports extensive and rich wildlife habitat, agriculture, and flood protection.
- Floods from the **Sacramento River**, operation of water infrastructure (**Big Notch**), and contributing watersheds (**Cache Creek**, **Colusa Basin via Ridgecut Slough**) affect the quantity and qualities of water sources entering the Bypass.
- Inputs interact with off-channel sites (Lower Elkhorn Basin Levee Set back project **YBLR4**, **SB4**; Tide's End Wetland **TEW**) to create habitat.
- Assessing the variable impact of water sources can help determine how flow and restoration contribute to habitat suitability.

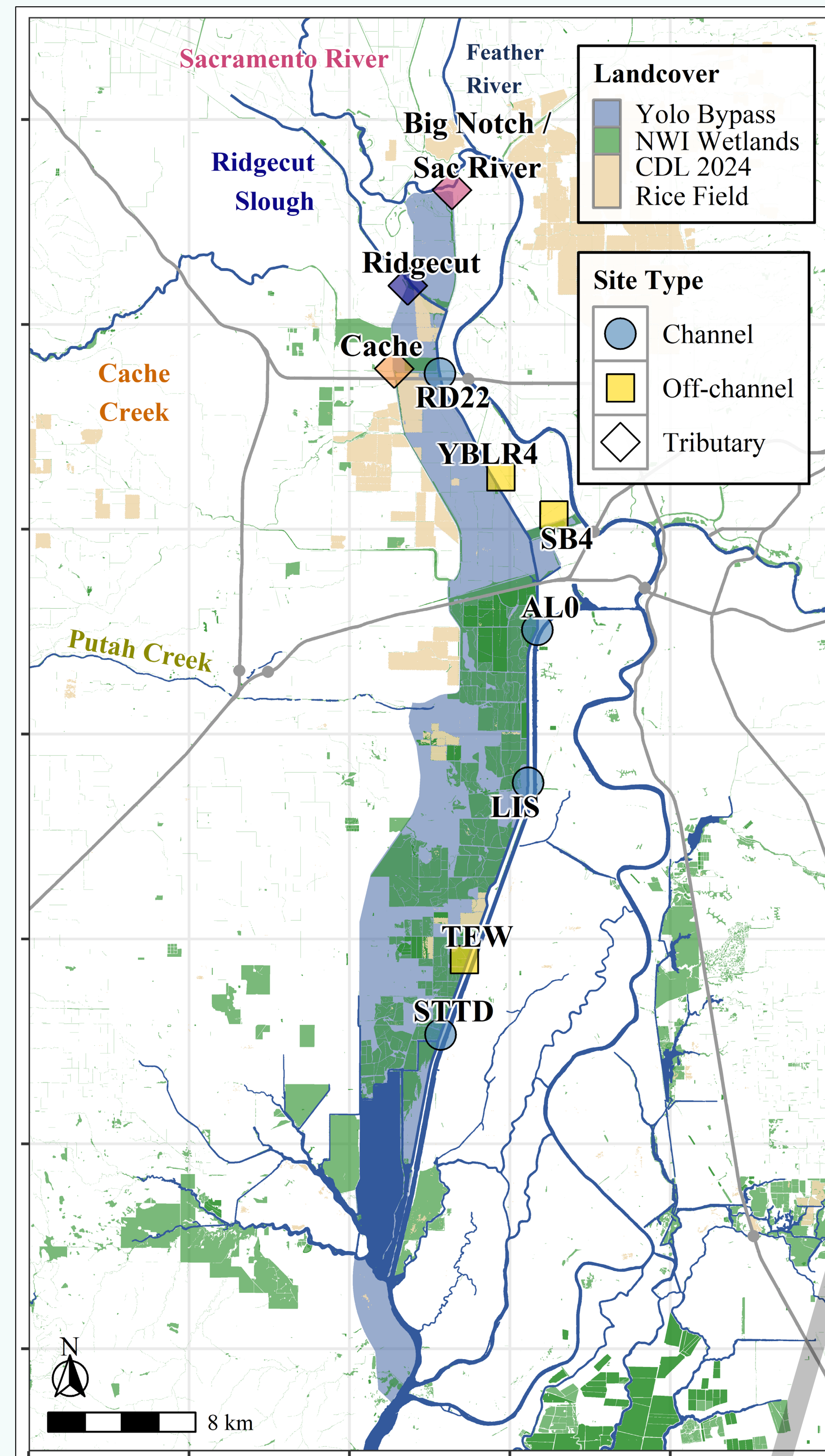
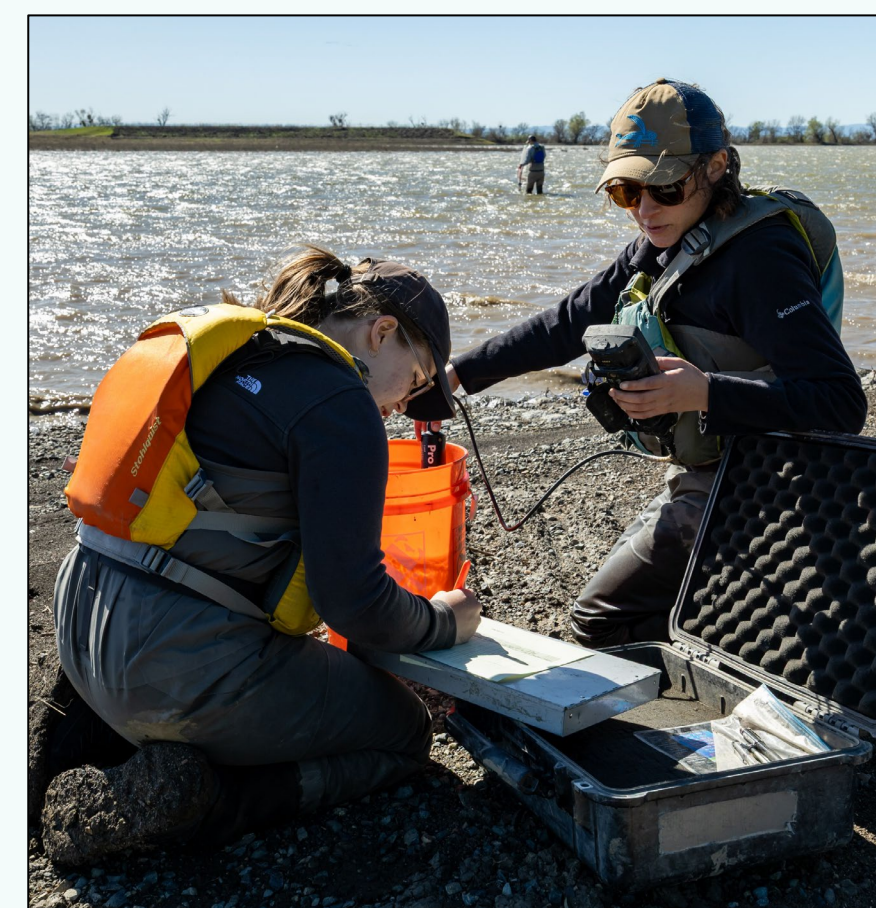


Figure 1. Map of the Yolo Bypass with sites sampled.

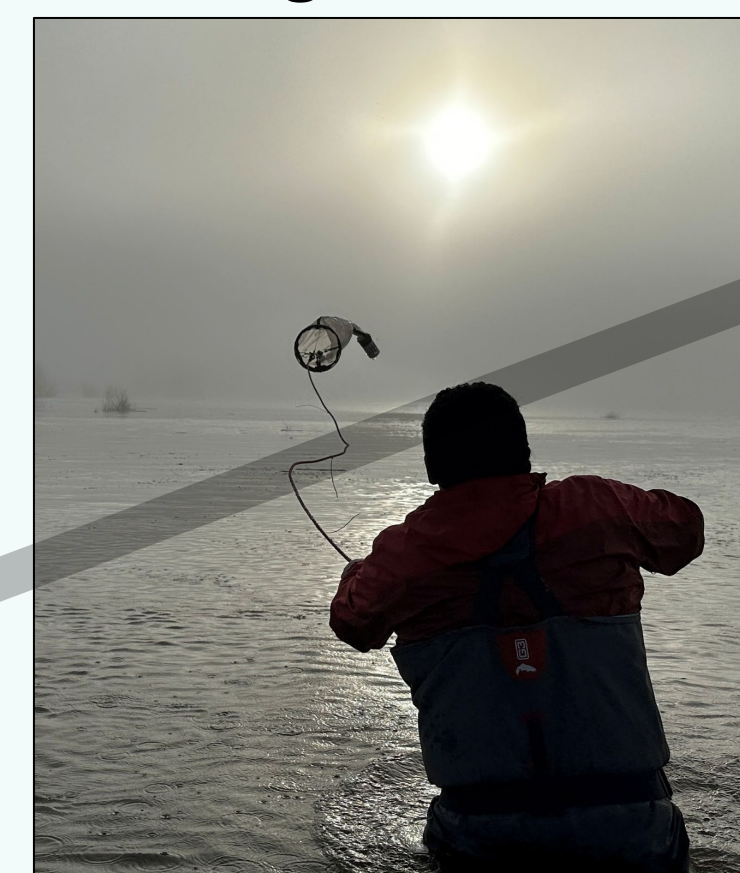
Weekly Sampling



- Point water quality (YSI EXO2 sonde)
- Specific conductivity (SPC, $\mu\text{S}/\text{cm}$)
 - Dissolved oxygen (DO, mg/L)
 - Fluorescent Dissolved Organic Matter (FDOM, QSU)
 - Turbidity (FNU)
 - Chlorophyll-a ($\mu\text{g}/\text{L}$)

Zooplankton (5 m net throws x4; 150 μm mesh; 30 cm ring)

- Quantified by taxa
 - Large cladocerans (e.g. *Daphnia pulex*) defined as >1mm length



Figures 2 and 3. Sampling water quality (left) and zooplankton (right).

Input Source and Quantity Influence Water Quality and Lower Trophic Conditions

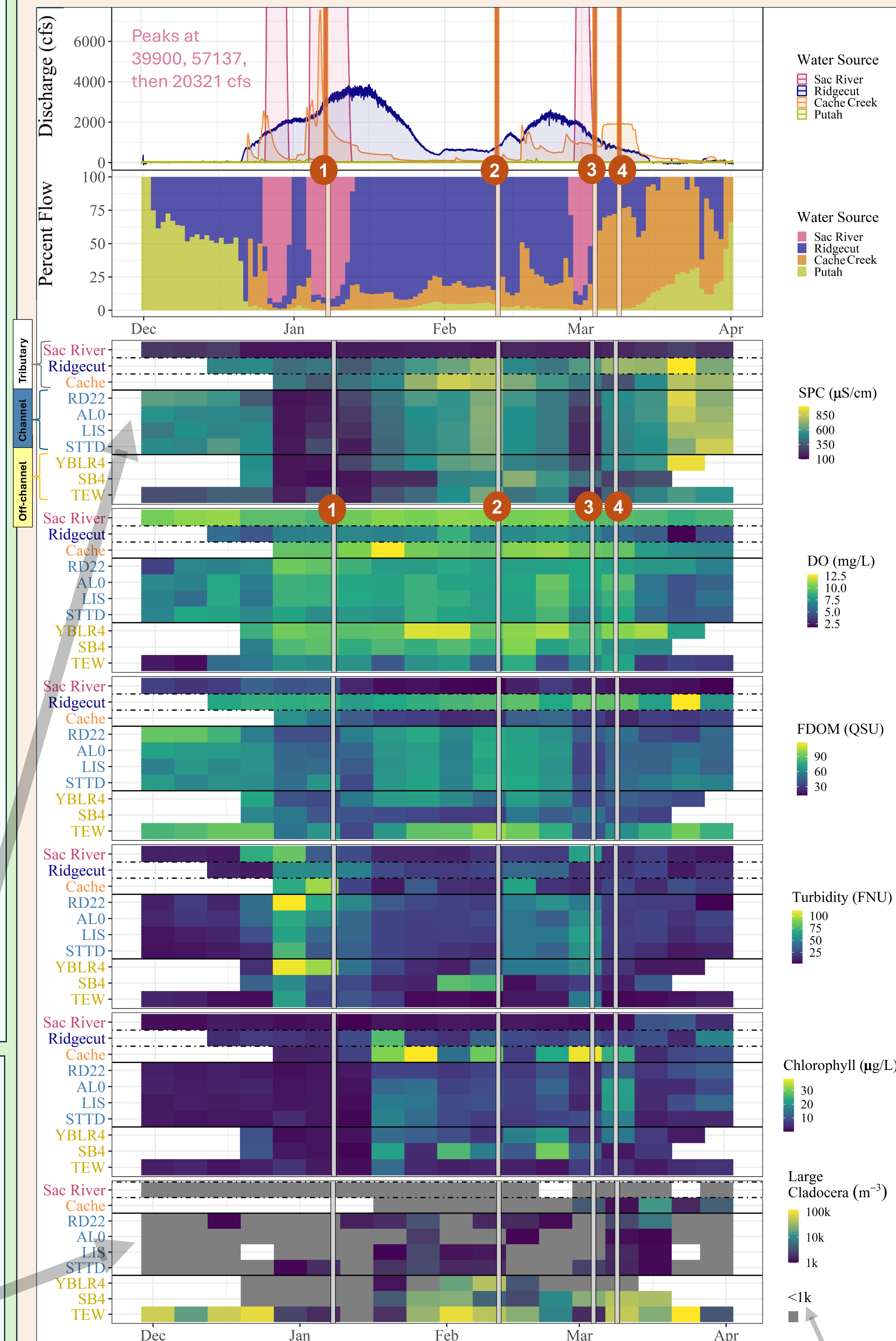
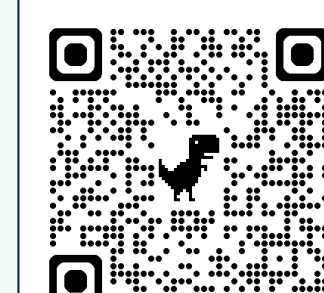


Figure 4. Tributary discharge (cfs) and percent flow (top). Heat maps of water quality parameters and large cladocera (zooplankton) density (bottom). Vertical bars match imagery dates.

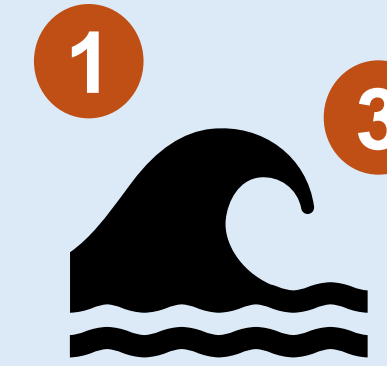
Optimal juvenile chinook growth@1k/m³ (Holmes et. al., In Review)

For more:
see [our dashboard!](#)



Acknowledgements: We would like to thank our Aquatic Ecology Unit and volunteers for helping collect and process the data: Matt Salvador, Anne Boyd, Lisa Vance, others. Thanks as well to DWR's North Central Region for providing sonde data and calibration. Our work is funded under the DWR State Water Project.

Sac River floods and suspends sediment



Ridgecut and Cache extend flood duration

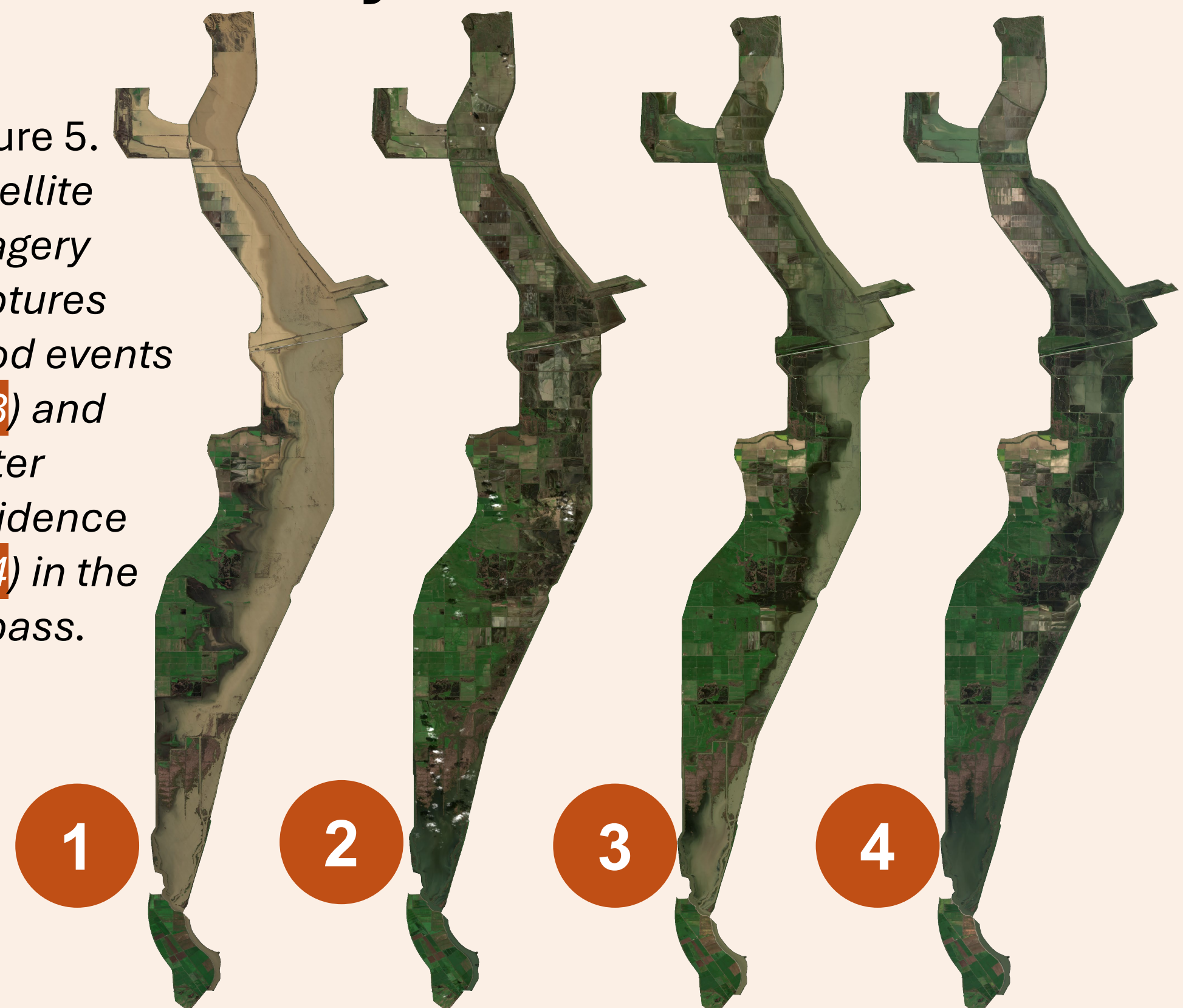


Water residence interacts with local features, differentiating habitat



Floodwaters Interact with Physical Habitat

Figure 5. Satellite imagery captures flood events (1,3) and water residence (2,4) in the Bypass.



Tributaries Display Distinct Water Signatures

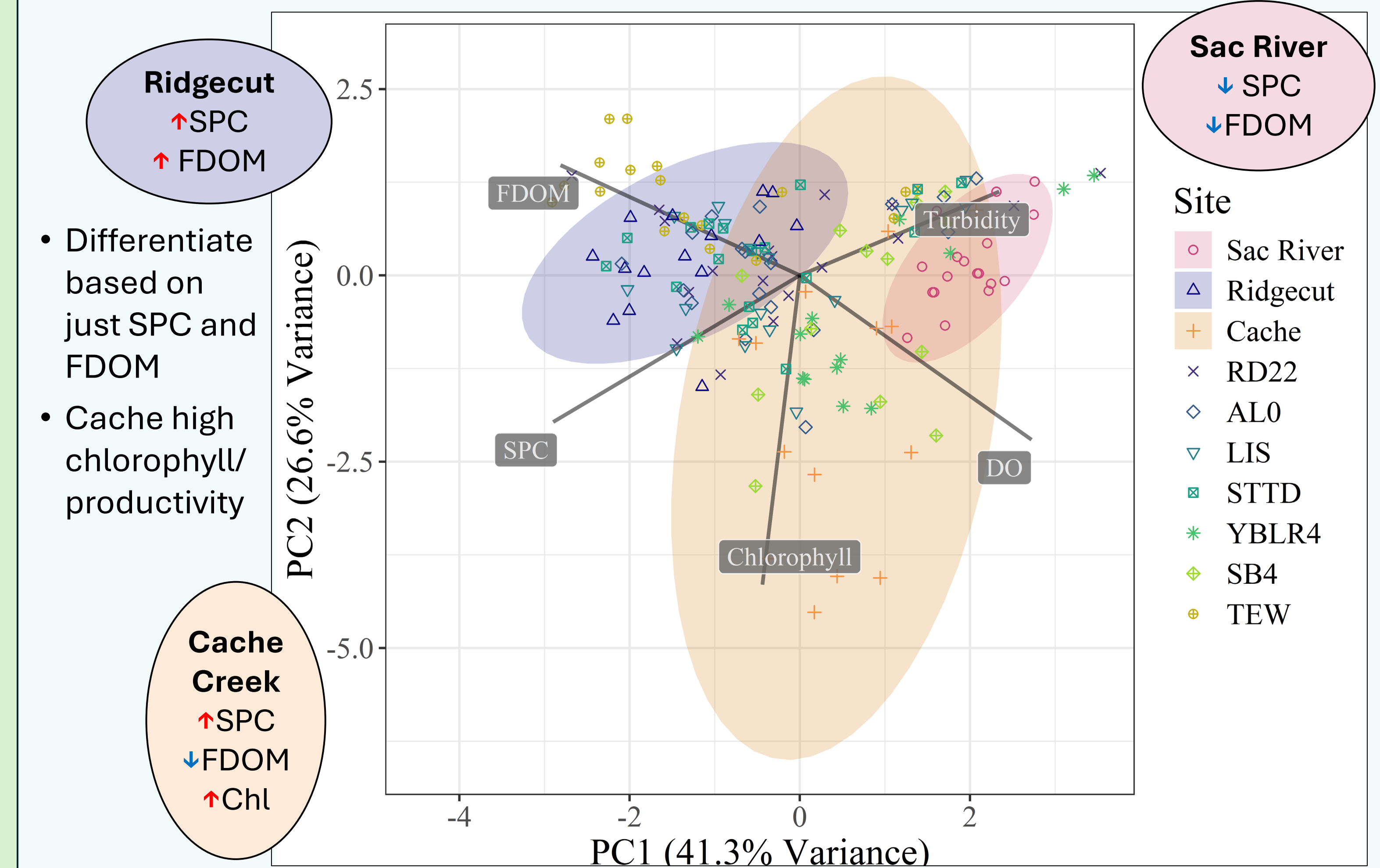


Figure 6. Principal component analysis (PCA) biplot of water quality with tributary ellipses.

Discussion and Implications

- Understanding contributions of tributaries is key to optimizing habitat suitability.
- Existing infrastructure controls tributary flows, which uniquely influence water quality, lower trophic conditions, and dispersion.
- Off-channel** floodplains acting as valuable habitat (rich in fish food) emphasizes land management considerations regarding agriculture and wetlands.
- Big Notch** connects the **Sac River** to the Bypass, promoting fish passage and nutrient delivery.
- Wallace Weir** and the **Knight's Landing Outfall Gates** control **Ridgecut** flow into the Bypass.
- Clear Lake** operations affect **Cache Creek** connectivity and flow to the Bypass.